ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

7zZWmnRuKu4KfuuJtvSb7H4uH8PUDq26QK8QY9TG81LJ

Date: 4/30/2025
Compounds: Amoxicillin, Paracetamol

New Compound Analysis

Name: Amoxacetol
Formula: $C_{16}H_{19}N_3O_5$
Weight: 333.34
Drug-likeness: 0.75

Activity:
Antimicrobial and Analgesic

Target Analysis

Bacterial Penicillin-binding proteins
Binding Affinity: -8.6
Key component in bacterial cell wall synthesis disruption

Cyclooxygenase enzymes (COX-1 and COX-2)
Binding Affinity: -8.2
Important for reducing pain and inflammation

Toxicity Profile

Risk Level: Moderate
Affected Systems: Liver, Kidneys

Mitigation Suggestions:
- Regular liver function tests
- Avoid in patients with hypersensitivity

Pharmacokinetics

Absorption:

Rapid gastrointestinal absorption

Distribution:
Widely distributed, crosses placenta, enters CNS

Metabolism:
Hepatic

Excretion:
Renal primarily, some biliary excretion

Half-life: 1.5 hours

Detailed Analysis

The creation of Amoxacetol results from the integration of the biochemical properties of Amoxicillin and Paracetamol. Amoxacetol is designed to target two critical pathways: bacterial cell wall synthesis inhibition and pain relief through COX enzyme inhibition. It features rapid absorption into systemic circulation and exhibits wide distribution, including CNS access. Hepatic metabolism and renal excretion are primary pharmacokinetic characteristics, with a moderate risk of liver and kidney toxicity. Clinical applications could extend to scenarios where simultaneous antibacterial and analgesic therapy is needed, leading to a streamlined treatment regimen. Drug design involved advanced computational techniques, predicting molecular interaction efficiencies and providing a framework for further synthesis and testing. Clinical phases will ascertain safety and efficacy, while continued research will address any unforeseen pharmacodynamic or pharmacokinetic challenges, to optimize therapeutic outcomes and minimizing adverse effects.
